

Code:

```
class Node:
```

```
    def __init__(self, data):
```

```
        self.data = data
```

```
        self.next = None
```

```
class Stack:
```

```
    def __init__(self):
```

```
        self.top = None
```

```
    def push(self, data):
```

```
        new_node = Node(data)
```

```
        new_node.next = self.top
```

```
        self.top = new_node
```

```
    def pop(self):
```

```
        if self.top is None:
```

```
            print("Stack Underflow")
```

```
            return None
```

```
        popped = self.top.data
```

```
        self.top = self.top.next
```

```
        return popped
```

```
    def peek(self):
```

```
        if self.top is None:
```

```
            print("Stack is empty")
```

```
        return None
    return self.top.data
```

```
def display(self):
    current = self.top
    while current:
        print(current.data, end=' ')
        current = current.next
    print()
```

```
stack = Stack()
```

```
stack.push(100)
```

```
stack.push(200)
```

```
stack.push(300)
```

```
print("Stack:", end=' ')
```

```
stack.display()
```

```
stack.pop()
```

```
stack.pop()
```

```
print("After Popping Twice:", end=' ')
```

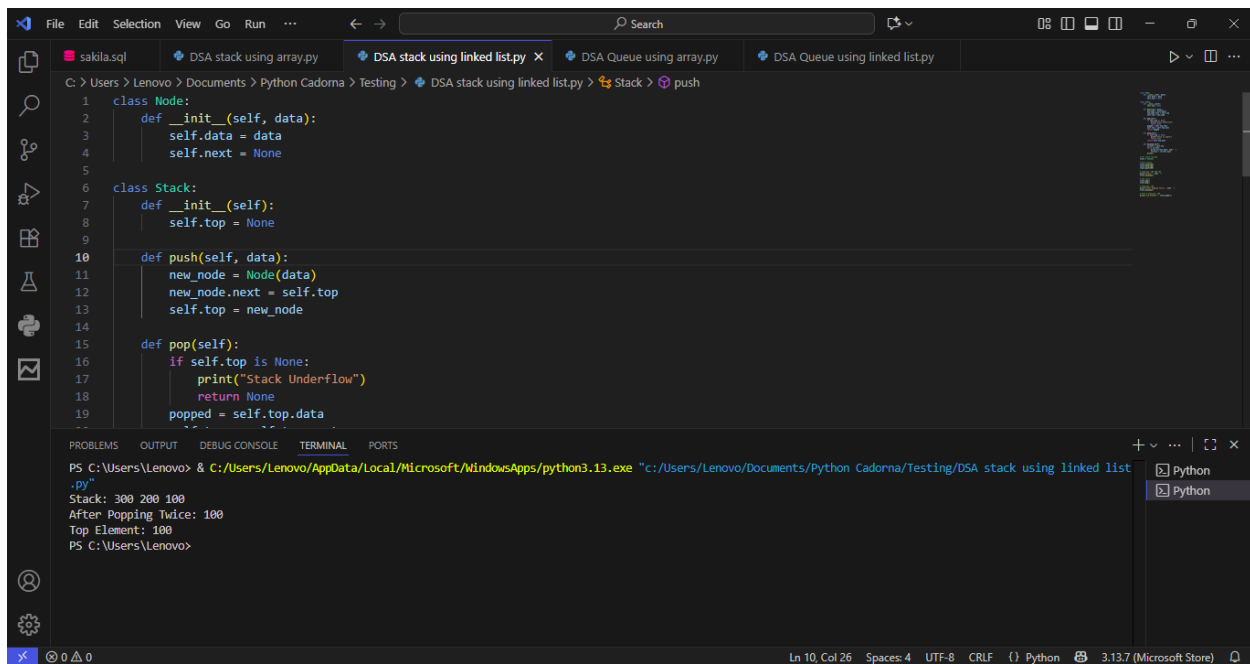
```
stack.display()
```

```
print("Top Element:", stack.peek())
```

Programiz Link:

<https://www.programiz.com/online-compiler/9xIK1UA4drDNY>

Sample Output:



```
1 class Node:
2     def __init__(self, data):
3         self.data = data
4         self.next = None
5
6 class Stack:
7     def __init__(self):
8         self.top = None
9
10    def push(self, data):
11        new_node = Node(data)
12        new_node.next = self.top
13        self.top = new_node
14
15    def pop(self):
16        if self.top is None:
17            print("Stack Underflow")
18            return None
19        popped = self.top.data
20        self.top = self.top.next
21        return popped
```

PS C:\Users\Lenovo> & C:/Users/Lenovo/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/Lenovo/Documents/Python Cadorna/Testing/DSA stack using linked list.py"

Stack: 300 200 100
After Popping Twice: 100
Top Element: 100
PS C:\Users\Lenovo>